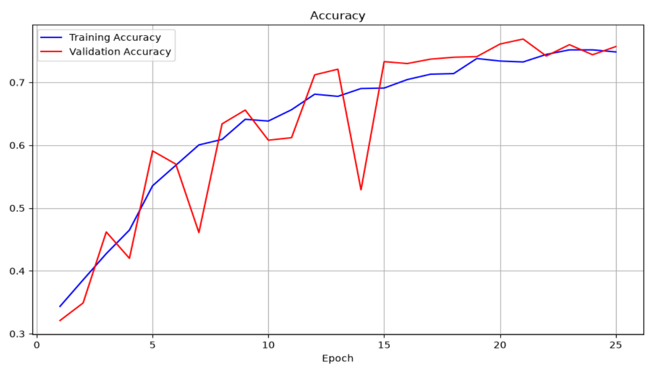
P6 Sections 5 & 6 — Facial Emotion CNN (FER2013)

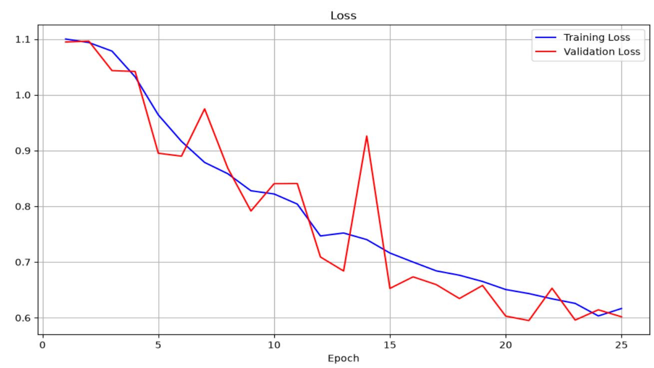
Dataset: Kaggle FER2013 (happy / neutral / surprise), 5000 training images via export_dataset.py.

Initial / Tuned Network Summary

Architecture: Rescaling → RandomFlip → RandomRotation → Conv(24)-Pool → Conv(32)-Pool → Conv(48)-Pool → Conv(64)-Pool → Dropout(0.25) → Flatten → Dense(32) → Dropout(0.4) → Softmax(3).
Total parameters: **149,683** (limit 150,000).
Epochs trained (with early stopping): **25**.
Best validation accuracy: **0.7690**.
Held-out test accuracy: **0.7561**, test loss: **0.6071**.
Saved model: **results/best_basic_model.keras**.

Training / Validation Curves



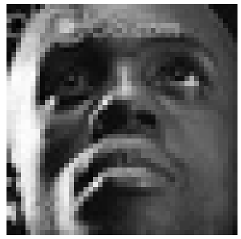


Hyperparameter Optimization Strategy (Section 6)

- Varied convolutional width (16–64 filters) while staying under 150k parameters.
- Tried Dense sizes 16 / 32 / 48; settled on Dense(32) for capacity without overfitting as quickly.
- Inserted 1–2 Dropout layers (rates 0.25 after last conv stack, 0.4 after Dense).
- Added light augmentation (horizontal flip + small rotation) to improve generalization.
- Lowered RMSprop learning rate from 0.001 to 0.0008.
- Used EarlyStopping on val_accuracy (patience=5) and ModelCheckpoint to keep the best epoch.
- Target: ≥60% for section 5 and ≥70% for section 6 on the held-out test set.

Example Training Images

neutral



happy

surprise